

Driver Package Instructions

You will first need to download the [driver package](#) from our archives and extract its contents at the root of an empty USB stick.

Installing the driver package

To install the driver package, simply plug the USB stick into the robot. The installation will start automatically. While the driver is installing, do not shut down the robot or remove the USB stick. Once the driver has installed, reboot the robot.

Concept of use

The driver package is a set of pre-made sub-programs that you have to call when you want to use them. They will be located in the folder “robotiq_2f_grippers_subprograms”. Example programs are available in this folder as well to show possible uses of the different sub-programs.

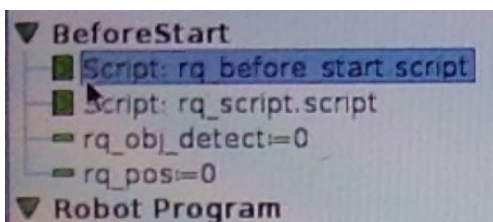
Before using the subprograms

You will need to add a script in the **Before Start** section of your program. When adding the script make sure that you choose to add it as a **File** instead of a line. Choose the script **rq_before_start.script**. Add a second script from a file. This second script will be **rq_script.script**.

Add two assignments, create two variables named like below and assign them to 0:

- rq_obj_detect
- rq_pos

Your **Before Start** section should look like this:



START PRODUCTION FASTER

When calling the subprograms, it will create a folder containing the instructions. You will need to rename the folder in this manner: **SubP_<name_of_the_subprogram>** otherwise, you will receive an error message.

Uninstalling the driver package

To uninstall the drivers, you will need to put the file **uninstall.sh**, located in the folder you downloaded from the archives and put it at the root of an empty USB stick. Plug this USB stick onto the robot and it will automatically remove all Robotiq 2-Finger Gripper drivers previously installed on the robot.

Subprograms definition

<code>rq_activate_and_wait()</code>	Activates the gripper then waits for the activation to be completed. (Recommended)
<code>rq_open_and_wait()</code>	Opens completely the gripper and waits for the action to be completed before doing the next instruction.
<code>rq_close_and_wait()</code>	Closes completely the gripper and waits for the action to be completed before doing the next instruction.
<code>rq_set_force(0-255)</code>	Expects an argument between 0 and 255. 0 being minimum force and 255 being maximum force.
<code>rq_set_speed(0-255)</code>	Expects an argument between 0 and 255. 0 being minimum speed and 255 being maximum speed.
<code>rq_is_motion_complete()</code>	Will return True if the gripper is at the expected position.
<code>rq_move_and_wait(0-255)</code>	Expects an argument between 0 and 255. 0 being completely opened, 255 is completely closed. Will wait for this action to be completed before doing the next instruction.
<code>rq_is_gripper_activated()</code>	Will return True if the gripper is activated.
<code>rq_is_object_detected()</code>	Will set the variable <i>rq_obj_detect</i> to 1 and return True if an object is currently detected, otherwise, it will set the variable <i>rq_obj_detect</i> to 0 and return False.
<code>rq_current_pos()</code>	Will assign the variable <i>rq_pos</i> to the current position of the gripper. This value should then be immediately cached into another variable.